

Aman Tiwari

Mobile: +91-7737102386
github.com/aman-tiwari18

Email: amantiwari18@gmail.com
linkedin.com/in/amantiwari18

EXPERIENCE

- Center for Developing Intelligent Systems (IIT Kanpur)** Kanpur, India
Software Engineer Oct 2025 – Present
 - Built a **consumer grievance analytics platform** over **4–5M complaint records**, delivering state-, company-, and category-wise distributions with **drill-down analytics** across category, subcategory, state, and company levels for government dashboards.
 - Designed high-performance backend services in **FastAPI** over **PostgreSQL** and **Elasticsearch**, using **ThreadPoolExecutor** concurrency and query optimization to reduce analytical API response time significantly.
 - Added caching for frequent analytics and **GPT-generated insights**, plus alerting to surface rising complaint trends and resolution-quality signals.
 - Developed interactive **React.js** dashboards (bar, pie, and donut charts) with optimized data fetching and modular components to visualize complaint trends.
 - Engineered a **large-scale intelligent video-surveillance system** for national examinations with real-time **RTSP/HLS** camera ingestion, architected to scale to **100,000+ (1 lakh) concurrent camera feeds** across a distributed multi-host GPU worker fleet.
 - Deployed **vision classification (vLLM)** and **object-detection (YOLO)** pipelines that classify camera scenes and detect exam-integrity events in real time, persisting detections in **PostgreSQL** for fast query and retrieval.
 - Built **React.js** monitoring dashboards for live camera status, event logs, and historical alerts, and optimized event-logging pipelines to reduce detection-to-dashboard latency.
 - Implemented real-time **push/SMS alerting**, shift-based detection scheduling, and job cancellation across the distributed fleet, with **Prometheus**-based GPU and system monitoring for reliability.
- Capsule** Remote
Full Stack Developer Jun 2024 – Jul 2025
 - Developed and maintained **RESTful APIs** using **Node.js** and **Express.js** for healthcare and pharmacy workflow systems.
 - Implemented **Role-Based Access Control (RBAC)** to secure backend services and enforce access policies.
 - Built **React.js dashboards and forms** integrated with backend APIs for operational data management.
 - Automated CI/CD pipelines using **Docker** and **GitHub Actions** for streamlined deployments.
- Linsible Technologies** Remote
Full Stack Developer Intern Feb 2024 – May 2024
 - Developed backend APIs using **Node.js**, **Express.js**, and **PostgreSQL** for healthcare applications.
 - Implemented caching using **Redis** and search capabilities with **Elasticsearch/OpenSearch** to improve data retrieval performance.

EDUCATION

- Bundelkhand Institute of Engineering and Technology** Jhansi, India
B.Tech in Electronics and Communication Engineering — *GPA: 7.30* 2019 – 2023

SKILLS

- Languages:** C++, JavaScript, TypeScript, Python
- Backend:** FastAPI, Node.js, Express.js, REST APIs, Authentication, RBAC
- Frontend:** React.js, Redux, HTML5, CSS3, Tailwind CSS
- Databases & Search:** PostgreSQL, MySQL, Elasticsearch, Redis
- AI & Vision:** vLLM, YOLO
- DevOps & Tools:** Git, GitHub, Docker, GitHub Actions, Linux, Prometheus
- Concepts:** Data Structures & Algorithms (500+ problems), System Design Basics, Operating Systems, DBMS, Networking

ACHIEVEMENTS

- Solved **500+ problems** on LeetCode.
- Secured **Rank 293** in GeeksforGeeks Mega Job-A-Thon.